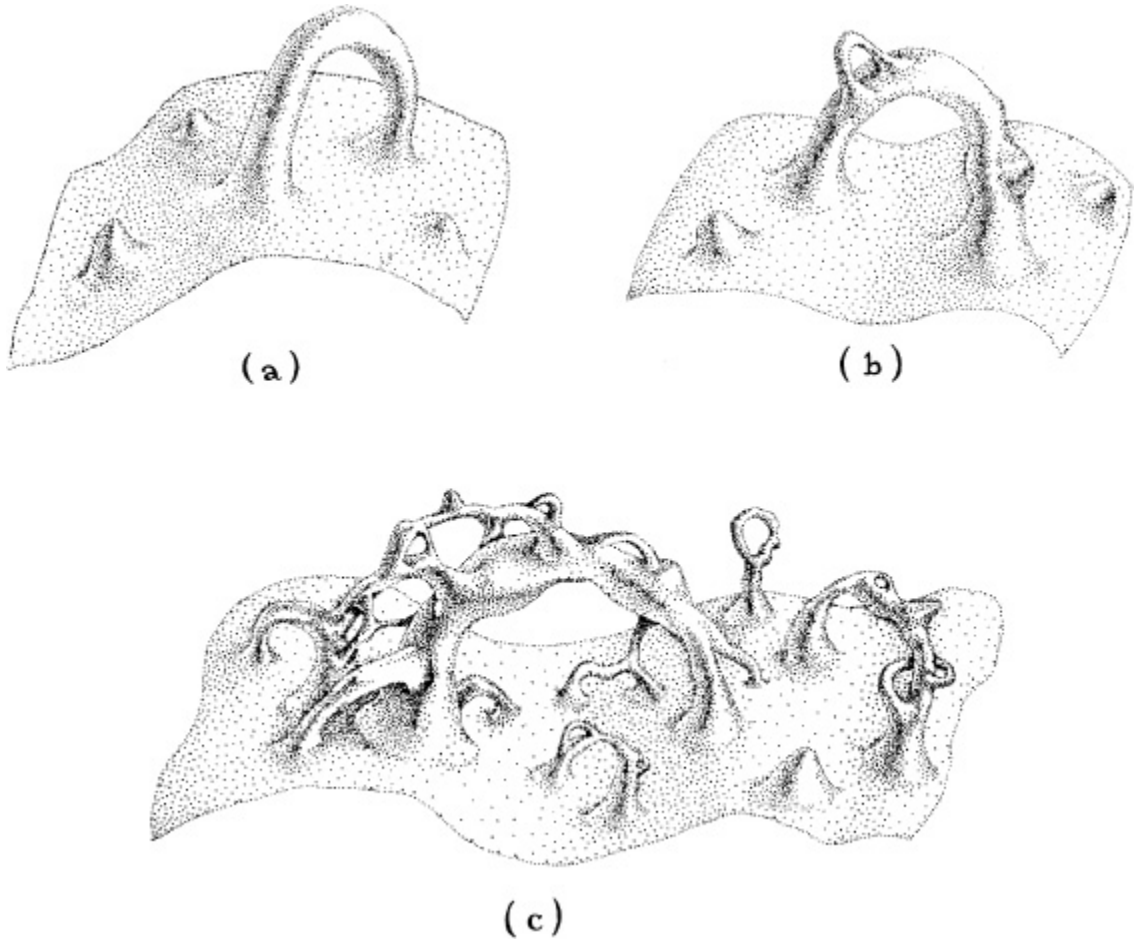
space would look completely smooth, with a very definite (but tiny) amount of curvature. As the microscopic zoom nears, then passes 10^{-32} centimeter, however, one would see space begin to writhe, ever so slightly at first, and then more and more strongly until, when a region just 10^{-33} centimeter in size fills the supermicroscope's entire eyepiece, space has become a froth of probabilistic quantum foam.



14.3 (Same as Figure 13.7.) Embedding diagrams illustrating quantum foam. The geometry and topology of space are not definite; instead, they are probabilistic. They might have, for example, a 0.1 percent probability for the form shown in (a), a 0.4 percent probability for (b), a 0.02 percent probability for (c), and so on.

Since the quantum foam is everywhere, it is tempting to imagine an infinitely advanced civilization reaching down into the quantum foam, finding in it a wormhole (say, the "big" one in Figure 14.3b with its 0.4 percent probability), and trying to grab that wormhole and enlarge it to